Measurement Map

Companion to the manuscript skeleton, 10 October 2026

1 Scope

The map takes each object of the model in `MANUSCRIPT.tex` to its counterpart in the Basel capital framework [Basel Committee on Banking Supervision, 2011], the Basel liquidity standard [Basel Committee on Banking Supervision, 2013] and the Call Report instructions for regulatory capital [Federal Financial Institutions Examination Council, 2026], and says how it is observed. The FDIC panel fields are defined in the FDIC variable workbook [Federal Deposit Insurance Corporation, 2026]. The regulatory cap is that of Bayraktar et al. [2026] (BCVW). Quotation and reading IDs refer to the `lit/` records, where each quotation is checked verbatim on its page. The rearrangements of the regulatory ratios are proved in `lean/mathlib/MeasurementMap.lean`.

A status of Rule means a regulation sets the value. Report means a regulatory report records the object for each bank. Calibration means the model takes a value with no regulatory counterpart. Unobserved means neither a rule nor a report gives it. Output means the model produces it and its observation is part of the empirical design.

2 Map

Table 1: Model objects and their observation.

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G1	x	The state, assets over liabilities.	Total assets over total liabilities, `ASSET/LIAB` in the FDIC panel. From the reported leverage ratio ℓ (Tier 1 over total assets for the leverage ratio), $x = 1/(1-\ell)$ holds only if Tier 1 equals book equity and the denominator equals total assets.	Report	`FDIC-Q1,FDIC-Q2,RCR-Q4,RCR-D3, MeasurementMap. leverage_of_state`
G2	π	The share of assets in the risky asset.	Risk-weighted assets over total assets, `RWAJT/ASSET`, under risk weights of 0 on the riskless and 1 on the risky asset. With a positive weight on the riskless asset the identification fails.	Report	`FDIC-Q3,RCR-Q5,RCR-D2,MeasurementMap. rwa_two_assets, MeasurementMap. control_rwa_needs_ zero_weight_on_ riskless`
G3	$X - L$	Equity, assets less liabilities.	CET1 capital or Tier 1 capital (`RBCT1J`) as reported. Both are net of regulatory adjustments, so they equal book equity only when the adjustments vanish.	Report	`FDIC-Q4,B3-Q3,RCR-Q1, RCR-Q3,B3-D2`
G4	a_1	The floor on equity over risky assets, the solvency branch $(x-1)/a_1$ of the cap.	The CET1 minimum of 4.5% of risk-weighted assets, which is BCVW's baseline 0.045, or the Tier 1 minimum of 6.0%. The bank's own ratio is reported as CET1 (`RBCT1CER`) or Tier 1 capital over risk-weighted assets. Banks electing the community bank leverage ratio report neither.	Rule	`FDIC-Q5,FDIC-Q6,BCVW-Q9,BCVW-Q13,B3-Q1,B3-Q2,B3-D1,RCR-Q2,RCR-Q6,RCR-Q7,RCR-D1, MeasurementMap. solvency_iff, MeasurementMap. control_solvency_ needs_positive_ exposure`

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G5	a_2	Net 30-day outflows as a share of liabilities, in the liquidity branch $(x-a_2)/a_3$ of the cap.	A run-off rate on all liabilities. BCVW's baseline 0.05 is the factor for stable insured retail deposits, and less stable retail deposits run off at 10% or more. The LCR must be at least 100%. No report read gives the rate.	Rule	`BCVW-Q10,LCR-Q1,LCR-Q9,LCR-Q10,LCR-D1, MeasurementMap.lcr_iff,MeasurementMap.control_lcr_needs_positive_runoff`
G6	a_3	The haircut on the risky asset in high-quality liquid assets, the slope $1/a_3$ of the liquidity branch.	The riskless asset counts in full, as Level 1 assets do. The standard's haircuts are 15% (Level 2A) and 25% or 50% (Level 2B). BCVW's baseline 0.30 is none of them. The 40% cap on Level 2 assets and the 75% cap on inflows are not in the model.	Calibration	`BCVW-Q11,BCVW-Q12, BCVW-Q13,LCR-Q2,LCR-Q3,LCR-Q4,LCR-Q5,LCR-Q6,LCR-Q7,LCR-Q8,LCR-D2,LCR-D3, MeasurementMap.hqla_haircut, MeasurementMap.cap_iff`
G7	q	The requirement on $x-1$, with $z=0$ at $x=1+q$.	A floor ℓ_0 on the leverage ratio is the floor $\ell_0/(1-\ell_0)$ on $x-1$, so the 3% Tier 1 leverage ratio gives $q=3/97$ when Tier 1 equals book equity. This is a reading, and the bundle does not fix q. The z-volatility does not depend on q.	Rule	`B3-Q7,RCR-Q4, MeasurementMap.leverage_req_iff, MeasurementMap.control_leverage_needs_l0_lt_one, MeasurementMap.zvol2_free_of_q`
G8	σ, μ_s, r	The volatility and drift of the risky asset and the riskless rate.	No regulatory counterpart. BCVW's baseline is $\sigma=0.08$, $\mu=0.04$, $r=0.02$.	Calibration	`BCVW-Q14`
G9	σ_L, c, γ, μ_L, r_L	The liability shock, its correlation with the risky return, the net income on liabilities, the liability drift and the deposit rate, with $\mu_L=\gamma+r_L$.	No regulatory counterpart. BCVW's baseline is $\sigma_L=0.03$, $c=0.20$, $\gamma=0.02$, $\mu_L=0.03$.	Calibration	`BCVW-Q14,M9`
G10	ρ_L	The liability-net discount rate of the reduced problem.	Formed from the shareholder discount rate, BCVW's baseline $\rho=0.12$. No regulatory counterpart.	Calibration	`BCVW-Q14`
G11	κ	The proportional issuance cost.	No regulatory counterpart. BCVW's baseline is $\kappa=0.01$ for dilution and $\kappa'=0.02$ per unit issued.	Calibration	`BCVW-Q14`
G12	K	The fixed issuance cost, charged per unit of liabilities.	Not observed. A cost fixed in dollars would break the reduction to x. Locally identified together with λ_S from the barrier and the trigger.	Unobserved	`BCVW-Q4,K2,M5,M6`
G13	λ_S	The asymmetry parameter of the shortfall penalty.	Not observed. The variance ratio carries no information about it. Locally identified together with K from the barrier and the trigger.	Unobserved	`K1,K2,M3,M6,M11`
G14	y^*	The dividend barrier.	The state at which the bank pays dividends (`EQCDIV`). Under the conservation buffer a bank with CET1 between 4.5% and 7.0% of risk-weighted assets has its distributions restricted, so observed payouts reflect the rule as well as the barrier.	Output	`FDIC-Q8,M4,B3-Q4,B3-Q5,B3-Q6,B3-D3`

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G15	x_L, y_{post}	The recapitalisation trigger and the injection target.	The state just before and just after a recapitalisation, measured as net stock issuance (EQCSXQ) plus capital from the parent holding company (EQCBHCTR), in quarters without merger-driven equity changes (EQCMRG).	Output	FDIC-Q9,FDIC-Q10, FDIC-Q12,FDIC-Q13,M5, M6,M7,M10
G16	ζ, v_1^2, v_3^2	The z-volatility and its limits at the distress boundary and deep in surplus.	The variance of changes in $z = \log((x-1)/q)$ near the distress boundary and deep in surplus. It does not depend on q.	Output	M2,M3,M8, MeasurementMap.zvol2_ free_of_q
G17	v_1^2/v_3^2	The deficit-to-surplus variance ratio, read in discrete time as λ_V^4.	The ratio of the two variances of G16. Reading it as λ_V^4 is a definition.	Output	K1,M1,M3
G18	z_t	The discrete-time state, the log of equity over required capital Q_t.	Tier 1 capital over 8.5% of risk-weighted assets, the Tier 1 minimum plus the conservation buffer.	Report	FDIC-Q3,FDIC-Q4,B3-Q2,B3-Q4,M12
G19	g	The growth rate of required capital, exogenous in the model.	Quarterly growth of risk-weighted assets.	Report	FDIC-Q3,M12
G20	b	The retention ratio.	One minus dividends over net income, over four quarters, defined when net income is positive.	Report	FDIC-Q7,FDIC-Q8,M12
G21	δ_p	The surplus payout rate.	The share of the log-surplus paid as dividends in a quarter.	Report	FDIC-Q8,M12
G22	ϕ^-, ϕ^+	Deficit and surplus persistence of z_t.	Least squares of z_t on z_{t-1} through the origin, on pairs starting in deficit or surplus.	Output	M12

3 What the empirical analysis measured

Each object of the map, and what the FDIC panel of US banks, 2015 to 2024, could do with it. Tracked means the object is measured and no model prediction about it is tested. Holds and Fails mean a prediction of the solved model about it is tested. Not measured means the data available do not measure the object. Each figure is printed by the computation in the cited results file, which the checker re-runs on its pinned data.

Table 2: What the data could measure.

ID	Measured as	What the data show	Model prediction	Outcome	Evidence
G1	ASSET/LIAB, 213,771 bank-quarters.	Median 1.1182, interquartile range 1.0997 to 1.1447.	None on its own.	Tracked	FDIC-Q1,FDIC-Q2, 06_empirical_ starter/results_ model_and_data.md
G2	RWAJT/ASSET, under the risk-weight identification.	Median 0.6857.	$\pi = \bar{\pi}$, 2.3794 at the median state.	Fails	FDIC-Q3, MeasurementMap.rwa_ two_assets,06_ empirical_starter/ results_model_and_ data.md
G3	Book equity over assets, and Tier 1 capital (RBCT1J).	Book equity over assets has median 10.57%, interquartile range 9.06% to 12.64%.	None on its own.	Tracked	FDIC-Q4,06_ empirical_starter/ results_model_and_ data.md

ID	Measured as	What the data show	Model prediction	Outcome	Evidence
G4	The bank's Tier 1 capital over risk-weighted assets.	Median 15.04%, 3.34 times a_1, and 2.78 times at the 25th percentile.	The ratio sits at a_1.	Fails	`FDIC-Q4,B3-Q1,B3-D1,06_empirical_starter/results_model_and_data.md`
G5	Not reported in the data read.	Nothing.	None tested.	Not measured	`G5`
G6	Not reported in the data read.	Nothing.	None tested.	Not measured	`G6`
G7	Not reported. The leverage ratio gives a candidate floor.	Nothing.	None tested.	Not measured	`G7`
G8	The volatility of book x is computed, which is not the economic σ.	Book volatility of x 0.01468 a year. The model's volatility of x is 0.06081 at the observed π, 4.1 times larger.	A level, on an economic-value basis.	Not measured	`06_empirical_starter/results_model_and_data.md`
G9	Not separately measured.	Nothing.	None tested.	Not measured	`G9`
G10	Not measured.	Nothing.	None tested.	Not measured	`G10`
G11	Not measured.	Nothing.	None tested.	Not measured	`G11`
G12	Through the trigger and target of G15, which the data do not show.	No trigger-and-target behaviour (G15).	Identified with λ_S from the thresholds.	Not measured	`K2,06_empirical_starter/results_distress_targeting.md`
G13	As G12.	As G12.	As G12.	Not measured	`K2,06_empirical_starter/results_distress_targeting.md`
G14	Quarters with a dividend (`EQCDIV`), by x.	56.2% of bank-quarters below $y^* = 1.5715$ pay a dividend. Payment falls as x rises, from 60.1% for x between 1.03 and 1.10 to 32.0% between 1.20 and 1.39.	No dividend below y^*.	Fails	`FDIC-Q8,M4,06_empirical_starter/results_dividend_location.md`
G15	Recapitalisations, x before and after, within bank, under the pre-registered rule.	Issue size is unrelated to the pre-issue state and the post-issue state rises with it, $b_{\text{post}} = +1.37$ and $+1.36$ for the two valid distress proxies. Issues start at a mean x of 1.11, against the trigger 1.03.	One target within a bank, $b_{\text{post}} = 0$.	Fails	`FDIC-Q9,FDIC-Q12,M5,06_empirical_starter/PREREG_distress_targeting.md,06_empirical_starter/results_distress_targeting.md`
G16	Not computed. A level would need the economic σ of G8.	Nothing.	v_1^2, v_3^2, which need the cap to bind (M3), and it does not (G2).	Not measured	`M3`
G17	Not computed on this panel.	Nothing.	v_1^2/v_3^2, under the same condition as G16.	Not measured	`K1,M3`
G18	Tier 1 capital over 8.5% of risk-weighted assets.	462 of 130,488 quarter pairs start in deficit, about 0.35%.	None on its own.	Tracked	`FDIC-Q3,FDIC-Q4,06_empirical_starter/results_persistence.md`
G19	Quarterly growth of risk-weighted assets.	Mean -0.47% a quarter on deficit pairs: banks in deficit shrink risk-weighted assets.	Exogenous, so no response to a deficit.	Fails	`FDIC-Q3,M12,06_empirical_starter/results_persistence.md`

ID	Measured as	What the data show	Model prediction	Outcome	Evidence
G20	One minus dividends over net income, four quarters.	0.3131 on the 108 deficit pairs where it is defined in $(0,1]$.	None on its own.	Tracked	`FDIC-Q7,FDIC-Q8, 06_empirical_ starter/results_ persistence.md`
G21	Share of the log-surplus paid out per quarter.	0.0211 a quarter.	None on its own.	Tracked	`FDIC-Q8,06_ empirical_starter/ results_ persistence.md`
G22	Within the pre-registered tests T1 to T3.	Deficit persistence 0.88 against 1.01 predicted (T1, fails). Surplus persistence 0.9959 against 0.9789 (T3, fails). The fall of ϕ^- in g is not measured (T2).	$\phi^- = 1-g(1-b)/b$, $\phi^+ = 1-\delta_p$.	Fails	`M12,06_empirical_ starter/PREREG_ persistence.md,06_ empirical_starter/ results_ persistence.md`

4 The ratio constraints in the state

With equity $X-L$, risk-weighted assets πX, high-quality liquid assets $X-a_3\pi X$ and net outflows $a_2 L$, the two ratio constraints of BCVW are the two branches of the cap in $x = X/L$. For $a_1, a_2, a_3, \pi, X, L > 0$,

$$a_1 \le \frac{X-L}{\pi X} \iff \pi x \le \frac{x-1}{a_1}, \qquad 1 \le \frac{X-a_3\pi X}{a_2 L} \iff \pi x \le \frac{x-a_2}{a_3},$$

and both hold if and only if $\pi x \le u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ (`MeasurementMap.solvency_iff`, `MeasurementMap.lcr_iff`, `MeasurementMap.cap_iff`). At $\pi = 0$ the left side of the first equivalence is undefined and the equivalence fails, and at $a_2 = 0$ the same holds for the second (the two controls). For $\ell_0 < 1$ and $X, L > 0$, $\ell_0 \le (X-L)/X \iff \ell_0/(1-\ell_0) \le x-1$ (`MeasurementMap.leverage_req_iff`).

5 Pending

- The CET1 ratio (`RBCT1CER`) is not yet in the panel. The corner test of G4 uses Tier 1 against the CET1 minimum (B3-D1).
- The unit of the FDIC amounts, thousands of dollars, is read from one API record and not from the workbook (FDIC-D1).
- Rows G1 to G3 assume regulatory capital equals book equity. Items 12 to 18 and 27 to 29 of Schedule RC-R, which give the adjustments, were not read.
- The tier that a_1 stands for is not fixed (B3-D1).

References

Basel Committee on Banking Supervision. Basel III: A global regulatory framework for more resilient banks and banking systems. Technical report, Bank for International Settlements, June 2011. December 2010, revised June 2011.

Basel Committee on Banking Supervision. Basel III: The liquidity coverage ratio and liquidity risk monitoring tools. Technical report, Bank for International Settlements, January 2013.

Erhan Bayraktar, Etienne Chevalier, Vathana Ly Vath, and Yuqiong Wang. Tractable bank capital structure: Optimal control under Basel III constraints, 2026. arXiv:2603.14557v2.

Federal Deposit Insurance Corporation. All financial reports: Variable definitions, BankFind Suite API. `https://api.fdic.gov/banks/docs`, 2026. Workbook "All Financial Reports.xlsx", sheet "Reference-Variables&Definitions".

Federal Financial Institutions Examination Council. *Instructions for Preparation of Consolidated Reports of Condition and Income (FFIEC 031 and 041), Schedule RC-R, Part I: Regulatory Capital Components and Ratios*. Federal Financial Institutions Examination Council, June 2026.